PROGRAMMING WITH PYTHON
SUBJECT CODE : 202044504

DEPARTMENT OF INFORMATION TECHNOLOGY
SEMESTER 5

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Github Link :
https://github.com/12502080603005-adit/12502080603005_Python_Assignment/tree/main/Assignment1

ASSIGNMENT-1

PROBLEM STATEMENT 1: You are given records of students in the form enrollment, name, semester, CPI and a list of subject marks. Build a program that stores the records using lists, tuples and dictionaries. For each semester, print the top K students by CPI. If CPI is the same, prefer the student with higher average marks; if still tied, prefer lexicographically smaller enrollment number. Also print the subject-wise topper for every subject code.

CODE :

```
def main():
    n, k, m = map(int, input().split())
    if not (1 <= n <= 100000):
        raise ValueError("Number of students must be between 1 and 100000.")

    if not (1 <= k <= 50):
        raise ValueError("K must be between 1 and 50.")

    if not (1 <= m <= 12):
        raise ValueError("Number of subjects must be between 1 and 12.")
    semester_students = {}
    subject_toppers = {}
    for i in range(1, m + 1):
        subject_toppers[f"S{i}"] = (-1, [])

    for _ in range(n):
        data = input().split()

        enrollment = data[0]
        name = data[1]
        semester = int(data[2])
        cpi = float(data[3])
        marks = list(map(int, data[4:]))

        if len(marks) != m:

            raise ValueError(
                f"{enrollment} must have exactly {m} subject marks."
            )

        if any(mark < 0 or mark > 100 for mark in marks):
            raise ValueError("Marks must be between 0 and 100.")
```

```
    if not (1 <= semester <= 8):
        raise ValueError("Semester must be between 1 and 8.")

    if not (0.0 <= cpi <= 10.0):
        raise ValueError("CPI must be between 0.0 and 10.0.")

    average_marks = sum(marks) / m

    student = (
        enrollment,
        name,
        cpi,
        average_marks,
        marks
    )

    if semester not in semester_students:
        semester_students[semester] = []
    semester_students[semester].append(student


    ) for i, mark in enumerate(marks, start=1):


        subject = f"S{i}"
        highest_mark, toppers = subject_toppers[subject]
        if mark > highest_mark:
            subject_toppers[subject] = (mark, [enrollment])

        elif mark == highest_mark:
            toppers.append(enrollment)
for semester in
    sorted(semester_students): students =
    semester_students[semester]
    students.sort(
        key=lambda student: (
            -student[2],
            -student[3],
            student[0]
        )
    )
    top_students = students[:k]
```

```
        enrollments = [student[0] for student in top_students]

        print(f"Semester {semester}:", *enrollments)

    for i in range(1, m + 1):
        subject = f"S{i}"

        highest_mark, toppers = subject_toppers[subject]

        toppers.sort()

        print(f"{subject}:",
*toppers) if _name_== "_main_":
    main()
```

OUTPUT:

```
5 2 3
2201 Asha 5 8.90 90 82 75
2202 Bharat 5 8.90 88 95 70
2203 Chaitra 3 9.10 91 88 92
2204 Dev 5 7.80 76 80 85
2205 Esha 3 9.10 91 90 88
Semester 3: 2203 2205
Semester 5: 2202 2201
S1: 2203 2205
S2: 2202
S3: 2203
```

PROBLEM STATEMENT 2: Given n candidate passwords, classify each as STRONG, WEAK_LENGTH, WEAK_PATTERN or COMPROMISED. A strong password must contain at least one lowercase letter, uppercase letter, digit and one special symbol from $#@. Its length must be between 6 and 12. It must not contain any dictionary word from a banned list as a contiguous substring, ignoring case. It must also not contain the same character more than 3 times consecutively.

CODE :

```
import re
class TrieNode:
    def __init__(self):
        self.children = {}
        self.is_end = False

class Trie:
    def __init__(self):
        self.root = TrieNode()
    def insert(self, word):
        node = self.root
        for char in word.lower():
            if char not in node.children:
                node.children[char] = TrieNode()

            node = node.children[char]

        node.is_end = True

    def contains_banned_word(self, password):

        password = password.lower()

        for start in range(len(password)):

            node = self.root

            for end in range(start, len(password)):
                char = password[end]
                if char not in node.children:
                    break

                node = node.children[char]
                if node.is_end:
```

```
                return Truereturn False

def has_required_characters(password):
    has_lower = False
    has_upper = False
    has_digit = False
    has_special = False

    for char in password:
        if char.islower():
            has_lower = True
        elif char.isupper():
            has_upper = True
        elif char.isdigit():
            has_digit = True
        elif char in "$#@":
            has_special = True

    return has_lower and has_upper and has_digit and has_special

def has_repeated_character(password):

    if not password:
        return False

    consecutive_count = 1

    for i in range(1, len(password)):
        if password[i] == password[i - 1]:
            consecutive_count += 1

            if consecutive_count > 3:
                return True
        else:
            consecutive_count = 1

    return False
def classify_password(password, banned_trie):
    if len(password) < 6 or len(password) > 12:
        return "WEAK_LENGTH"

    if not has_required_characters(password):
        return "WEAK_PATTERN"
```

```
    if has_repeated_character(password):
        return "WEAK_PATTERN"

    if banned_trie.contains_banned_word(password):
        return "COMPROMISED"
return
"STRONG" def
main():
    b = int(input().strip())
    if not (1 <= b <= 10000):
        raise ValueError("Number of banned words must be between 1 and 10000.")
    banned_trie = Trie()

    for _ in range(b):
        word = input().strip()
        if word:
            banned_trie.insert(word)
    n = int(input().strip())

    if not (1 <= n <= 100000):
        raise ValueError("Number of passwords must be between 1 and 100000.")

    for index in range(1, n + 1):
        password = input().rstrip("\n")
        result = classify_password(password, banned_trie)
        print(f"{index}: {result}")

if _name____== "_main_":
    main()
```

OUTPUT:

```
2
admin
college
5
Abc12@
1: STRONG
adminA1@
2: COMPROMISED
AAAA1@b
3: WEAK_PATTERN
short
4: WEAK_LENGTH
Good9#
5: STRONG
```

PROBLEM STATEMENT 3: Implement a recursive expression evaluator for integer expressions containing non-negative integers, +, -, *, parentheses and variable references. Each variable may be defined by another expression. Detect cyclic variable dependencies and report CYCLE. Use memoization so that repeated variable evaluations are not recomputed.

CODE :

```
import re
class ExpressionEngine:
    def __init__(self, variables):
        self.variables = variables
        self.memo = {}
        self.visiting = set()
        self.variable_pattern = re.compile(r"^[A-Za-z_][A-Za-z0-9_]*$")

    def tokenize(self, expression):

        tokens = []
        i = 0

        while i < len(expression):

            if
                expression[i].isspace()
                : i += 1
                continue

            if expression[i].isdigit():
                j = i

                while j < len(expression) and
                    expression[j].isdigit(): j += 1

                tokens.append(("NUMBER", expression[i:j]))
                i = j
                continue

            if expression[i].isalpha() or expression[i] == "_":
                j = i

                while (
                    j < len(expression)
                    and (expression[j].isalnum() or expression[j] == "_")
                ):
```

```
            j += 1
        tokens.append(("VARIABLE", expression[i:j]))i = j continue

    if expression[i] in "+-*()":
        tokens.append((expression[i], expression[i]))
        i += 1
        continue

    raise ValueError("Invalid character in expression.")

  return tokens

def evaluate_variable(self, name):

  if name in self.memo:
    return self.memo[name]

  if name not in self.variables:
    raise ValueError("Unknown variable.")

  if name in self.visiting:
    raise RuntimeError("CYCLE")

  self.visiting.add(name)

  try:
    tokens = self.tokenize(self.variables[name])
    value, position = self.parse_expression(tokens, 0)
    if position != len(tokens):
      raise ValueError("Invalid expression.")

    self.memo[name] = value

    return value

  finally:
    self.visiting.remove(name)

def parse_expression(self, tokens, position):

  value, position = self.parse_term(tokens, position)
  while position < len(tokens):
```

```
        operator = tokens[position][0]
        if operator not in ("+", "-"):
            break

        position += 1

        right_value, position = self.parse_term(tokens, position)

        if operator == "+":
            value += right_value
        else:
            value -= right_value

    return value, position

def parse_term(self, tokens, position):

    value, position = self.parse_factor(tokens, position)

    while position < len(tokens):

        operator = tokens[position][0]

        if operator != "*":
            break

        position += 1

        right_value, position = self.parse_factor(tokens, position)

        value *= right_value

    return value, position
def parse_factor(self, tokens, position):
    if position >= len(tokens):
        raise ValueError("Invalid expression.")

    token_type, token_value = tokens[position]

    if token_type == "NUMBER":
```

```
            return int(token_value), position + 1

        if token_type == "VARIABLE":

            if token_value not in self.variables:
                raise ValueError("Unknown variable.")

            value = self.evaluate_variable(token_value)

            return value, position + 1

        if token_type == "(":

            value, position = self.parse_expression(
                tokens,
                position + 1
            )

            if (
                position >= len(tokens)
                or tokens[position][0] != ")"
            ):
                raise ValueError("Missing closing parenthesis.")
            return value, position + 1
        raise ValueError("Invalid expression.")
    def evaluate(self, expression):
        tokens = self.tokenize(expression)
        if not tokens:
            raise ValueError("Empty expression.")
        value, position = self.parse_expression(tokens, 0)

        if position != len(tokens):
            raise ValueError("Invalid expression.")
        return value

def main():
    v = int(input().strip())

    if v < 1:
        raise ValueError("Number of variables must be at least 1.")
    variables = {}
    for _ in range(v):
```

```
        line = input().strip()
        if "=" not in line:
            print("INVALID")
            return

        name, expression = line.split("=", 1)
        name = name.strip()
        expression = expression.strip()
        if not re.match(r"^[A-Za-z_][A-Za-z0-9_]*$", name):
            print("INVALID")
            return
        if not expression:
            print("INVALID")
            return
        variables[name] = expression
    final_expression = input().strip()

    try:
        engine = ExpressionEngine(variables)
        result =
        engine.evaluate(final_expression)
        print(result)

    except RuntimeError as error:
        if str(error) == "CYCLE":
            print("CYCLE")
        else:
            print("INVALID")
    except (ValueError, RecursionError):
        print("INVALID")

if __name__ == "__main__":
    main()
```

OUTPUT:

```
3
a = 2 + 3
b = a * 4
c = b - (a + 1)
c + 10
24
```

PROBLEM STATEMENT 4: A CSV file contains transaction_id, account_id, type, amount and timestamp. Create a program that reads the file, validates every row, writes valid credit and debit transactions into separate CSV files, writes rejected rows into error.csv with reason, and finally prints account-wise balance changes. The program must continue processing after invalid rows using exception handling.

CODE :

```
import csv
from datetime import datetime

def validate_transaction(row):
    required_fields = ["tid", "acc", "type", "amount", "time"]

    for field in required_fields:
        if field not in row or not row[field].strip():
            return False, f"Missing {field}"

    transaction_type = row["type"].strip().upper()

    if transaction_type not in ("CREDIT", "DEBIT"):
        return False, "Invalid transaction type"

    try:
        amount = float(row["amount"])
    except ValueError:
        return False, "Amount is not numeric"

    if amount <= 0:
        return False, "Amount must be greater than zero"

    try:
        datetime.strptime(
            row["time"].strip(),
            "%Y-%m-%dT%H:%M:%S"
        )
    except ValueError:
        return False, "Invalid timestamp"

    return True, ""

def

    process_file(filename):
```

```
 balances = {}try:

with open(filename, "r", newline="", encoding="utf-8") as infile:

    reader = csv.DictReader(infile)

    required = {"tid", "acc", "type", "amount", "time"}

    if not reader.fieldnames or not required.issubset(
        set(reader.fieldnames)
    ):
        print("Invalid CSV header.")
        return

    with open(
        "credit.csv", "w", newline="", encoding="utf-8"
    ) as credit_file, \
    open(
        "debit.csv", "w", newline="", encoding="utf-8"
    ) as debit_file, \
    open(
        "error.csv", "w", newline="", encoding="utf-8"
    ) as error_file:

        credit_writer = csv.DictWriter(
            credit_file,
            fieldnames=reader.fieldnames
        )

        debit_writer = csv.DictWriter(
            debit_file,
            fieldnames=reader.fieldnames
        )
        error_fields = reader.fieldnames + ["reason"]


        error_writer = csv.DictWriter(


            error_file,
            fieldnames=error_fields
        )
```

```
            credit_writer.writeheader()
            debit_writer.writeheader()
            error_writer.writeheader()f
            or row in reader: try:
                valid, reason = validate_transaction(row)

                if not valid:
                    row_with_error = row.copy()
                    row_with_error["reason"] = reason
                    error_writer.writerow(row_with_error)
                    continue

                account = row["acc"].strip()
                transaction_type = row["type"].strip().upper()
                amount = float(row["amount"])

                balances.setdefault(account, 0)

                if transaction_type == "CREDIT":
                    credit_writer.writerow(row)
                    balances[account] += amount

                else:
                    debit_writer.writerow(row)
                    balances[account] -= amount

            except Exception as error:
                row_with_error = row.copy()
                row_with_error["reason"] = str(error)
                error_writer.writerow(row_with_error
                )

    except FileNotFoundError:
        print("Input file not found.")
        return
    except Exception as error:
        print("Error:", error) return
    sorted_balances = sorted(
        balances.items(),
        key=lambda item: abs(item[1]),
        reverse=True
    )
```

```
    print("Account-wise balance changes:")
    for account, balance in sorted_balances:
        if balance.is_integer():
            print(account, int(balance))
        else:
            print(account, balance)

    print("Files created: credit.csv, debit.csv, error.csv")

def main():

    filename = input("Enter CSV file path: ").strip()

    if not filename:
        print("File path cannot be empty.")
        return

    process_file(filename)

if __name__ == "__main__":
    main()
```

OUTPUT:

```
Enter CSV file path: transactions.csv
Account-wise balance changes:
A1 500
A2 -100
Files created: credit.csv, debit.csv, error.csv
```

PROBLEM STATEMENT 5: Design an OOP-based settlement engine with Account, Transaction and Bank classes. The engine must support deposit, withdraw and transfer operations, maintain transaction history, prevent overdraft, and rollback an entire batch if any transaction in the batch fails. Use custom exceptions and suitable getters/setters or properties.

CODE :

```
class BankError(Exception):
    pass
class
    AccountNotFoundError(BankError):
    pass
class InvalidAmountError(BankError):
    pass
class InsufficientBalanceError(BankError):
    pass
class Account:
    def __init__(self, account_id, balance=0):
        self.account_id = account_id
        self._balance = balance
        self._history = []

    @property
    def balance(self):
        return self._balance
    def deposit(self, amount):
        if amount <= 0:
            raise InvalidAmountError("Amount must be positive.")
        self._balance += amount

    def withdraw(self, amount):
        if amount <= 0:
            raise InvalidAmountError("Amount must be positive.")
        if amount > self._balance:
            raise InsufficientBalanceError(
                f"Insufficient balance in {self.account_id}"
            )
        self._balance -= amount

class Transaction:

    def __init__(self, operation, account_from=None,
                 account_to=None, amount=0):
        self.operation = operation
        self.account_from = account_from
```

```
        self.account_to = account_to
        self.amount = amountdef __str
        (self):
        return (
            f"{self.operation}: "
            f"{self.account_from} -> {self.account_to} "
            f"{self.amount}"
        )

class Bank:
    def __init__(self):
        self.accounts = {}
        self.history = []
    def add_account(self, account_id, balance):

        if balance < 0:
            raise InvalidAmountError(
                "Initial balance cannot be negative."
            )
        self.accounts[account_id] = Account(
            account_id,
            balance
        )

    def get_account(self, account_id):
        if account_id not in self.accounts:
            raise AccountNotFoundError(
                f"Account {account_id} does not exist."
            )

        return self.accounts[account_id]
    def deposit(self, account_id,
    amount):
        account = self.get_account(account_id)
        account.deposit(amount)
        transaction = Transaction(
            "DEPOSIT",
            account_to=account_id,
            amount=amount
        )
```

```
        self.history.append(transaction)

    def withdraw(self, account_id, amount):
        account = self.get_account(account_id)

        account.withdraw(amount)
        transaction = Transaction(
            "WITHDRAW",
            account_from=account_id,
            amount=amount
        )

        self.history.append(transaction)

    def transfer(self, from_id, to_id, amount):
        source = self.get_account(from_id)
        destination = self.get_account(to_id)
        source.withdraw(amount)
        destination.deposit(amount)

        transaction = Transaction(
            "TRANSFER",
            account_from=from_id,
            account_to=to_id,
            amount=amount
        )
        self.history.append(transaction)
    def snapshot(self):
        return {
            account_id: account.balance
            for account_id, account in self.accounts.items()
        }
    def restore(self, snapshot):
        for account_id, balance in snapshot.items():
            self.accounts[account_id]._balance = balance

def main():
    bank = Bank()
    n = int(input().strip())
    for _ in range(n):
        account_id, balance = input().split()
```

```
        bank.add_account(
            account_id,
            int(balance))

    q = int(input().strip())

    batch_number = 0
    batch_snapshot = None
    failed_batches = []

    for _ in range(q):
        parts = input().split()

        if not parts:
            continue

        operation = parts[0]

        if operation == "BATCH_BEGIN":

            batch_number += 1

            batch_snapshot = bank.snapshot()
            continue

        if operation == "BATCH_END":
            batch_snapshot = None

            continue

        try:

            if operation == "DEPOSIT":
                account_id = parts[1]
                amount = int(parts[2])
                bank.deposit(account_id,  amount)

            elif operation == "WITHDRAW":
                account_id = parts[1]
                amount = int(parts[2])
                bank.withdraw(account_id, amount)
            elif operation == "TRANSFER":
```

```
            from_id = parts[1]


                to_id = parts[2]amount = int(parts[3]) bank.transfer(
                    from_id
                    , to_id,
                    amount
                )
            else:
                raise BankError("Unknown operation.")
        except BankError:
            if batch_snapshot is not None:
                bank.restore(batch_snapshot)
                failed_batches.append(batch_number)

                while True:
                    remaining = input().split()
                    if remaining and remaining[0] == "BATCH_END":
                        break
                batch_snapshot = None
    for number in failed_batches:
        print(f"FAILED {number}")
    for account_id in sorted(bank.accounts):
        balance =
        bank.accounts[account_id].balance
        print(account_id, balance)

if _name____== "_main_":
    main()
```

OUTPUT:

```
3
A 1000
B 200
C 0
5
TRANSFER A B 300
BATCH_BEGIN
WITHDRAW B 1000
DEPOSIT C 50
BATCH_END


FAILED 1
A 700
B 500
C 0
```

PROBLEM STATEMENT 6: Given a set of Python module names and import relationships, determine a valid loading order. If a circular dependency exists, print CYCLE and the modules involved in one cycle. Ignore duplicate import edges. If multiple modules can be loaded, choose lexicographically smallest module first.

CODE :

```
import heapq
def main():
    n, e = map(int, input().split())
    modules = []

    for _ in range(n):
        modules.append(input().strip())

    adjacency = {
        module: [] for module in modules
    }

    indegree = {
        module: 0 for module in modules
    }

    seen_edges = set()

    for _ in range(e):

        module_a, module_b = input().split()

        edge = (module_b, module_a)

        if edge in seen_edges:
            continue

        seen_edges.add(edge)

        adjacency[module_b].append(module_a)
        indegree[module_a] += 1

    heap = []

    for module in modules:
        if indegree[module] == 0:
            heapq.heappush(heap,
```

```
module)order = [] while heap:

current =

heapq.heappop(heap)

order.append(current)

for dependent in adjacency[current]:
    indegree[dependent] -= 1

    if indegree[dependent] == 0:
        heapq.heappush(heap, dependent)

if len(order) != n:

print("CYCLE")

state = {module: 0 for module in modules}
parent = {}
cycle = []
def dfs(node):

    state[node] = 1

    for next_node in adjacency[node]:

        if state[next_node] == 0:

            parent[next_node] = node

            if
                dfs(next_node)
                : return True

        elif state[next_node] == 1:

            cycle_start = next_node
            cycle.append(cycle_start)
            current = node
            while current != cycle_start:
            cycle.append(current) current =
            parent[current]
```

```
                cycle.append(cycle_start)
                cycle.reverse()
                return True

            state[node] = 2
            return False

        for module in modules:
            if state[module] == 0:

                if dfs(module):
                    break
        print("
    ".join(cycle)) else:
        print(" ".join(order))

if _name____== "_main_":
    main()
```

OUTPUT:

```
4 3
core
db
ui
auth
ui auth
auth db
db core
core db auth ui
```

PROBLEM STATEMENT 7: Build an interactive calculator that accepts formulas of the form operand operator operand. Operands may be integers, decimals or previously stored variables. Supported operators are +, -, *, / and %. Invalid format, unknown variables, division by zero and unsupported operators must raise separate custom exceptions. The calculator must continue until the user enters quit.

CODE :

```
class
   InvalidFormatError(Exception):
   pass
class UnknownVariableError(Exception):
   pass
class DivisionByZeroError(Exception):
   pass
class
   UnsupportedOperatorError(Exception):
   pass
def get_operand(value, variables):
   try:
      return float(value)
   except ValueError:
      pass

   if value in variables:
      return
      variables[value]

   raise UnknownVariableError(
      f"Unknown variable: {value}"
   )

def calculate(left, operator, right):
   if operator == "+":
      return left + right

   if operator == "-":
      return left - right

   if operator == "*":
      return left * right

   if operator == "/":
```

```
        if right == 0:
            raise DivisionByZeroError(
                "Cannot divide by zero."
            )

        return left / right

    if operator == "%":

        if right == 0:
            raise DivisionByZeroError(
                "Cannot use modulo with zero."
            )
        return left % right

    raise UnsupportedOperatorError(
        f"Unsupported operator: {operator}"
    )

def format_result(value):
    if value.is_integer():
        return str(int(value))

    return str(value)

def main():

    variables = {}

    print("Interactive Calculator")
    print("Enter 'quit' to exit.")

    while True:

        try:

            line = input().strip()

            if line.lower() == "quit":
                break

            if not line:
```

```
        raise InvalidFormatError(
            "Empty input."
        )

    if "=" in line:
        parts = line.split("=")
        if len(parts) != 2:
            raise InvalidFormatError(
                "Invalid assignment format."
            )
        variable = parts[0].strip()
        value_text = parts[1].strip()

        if not variable.isidentifier():
            raise InvalidFormatError(
                "Invalid variable name."
            )
            if not value_text:
            raise
                InvalidFormatError(
                "Missing value."
            )

        value = get_operand(
            value_text,
            variables
        )
        variables[variable] = value
        continue
    parts = line.split()
    if len(parts) != 3:
        raise InvalidFormatError(
            "Formula must be: operand operator operand"
        )
    left_text, operator, right_text = parts

    left = get_operand(
        left_text,
        variables
    )

    right = get_operand(
```

```
        right_text,variables
        )

        result = calculate(
            left,
            operator,
            right
        )

        print(format_result(result))

    except DivisionByZeroError as error:
        print("DivisionByZeroError")
        print(error)
except UnsupportedOperatorError:
        print("UnsupportedOperatorError")

 except UnknownVariableError as error:
        print("UnknownVariableError")
        print(error)

except     InvalidFormatError     as
        error:
        print("InvalidFormatError"
        ) print(error)
if _name_== "_main_": main()
```

OUTPUT:

```
Interactive Calculator
Enter 'quit' to exit.
x = 10
x + 5
15
x / 0
DivisionByZeroError
Cannot divide by zero.
3 ** 2
UnsupportedOperatorError
quit
```

PROBLEM STATEMENT 8: A server produces several text log files. Create a program that scans all files in a folder, builds an inverted index mapping each normalized token to the list of file names and line numbers where it appears, stores the index using pickle, and compresses both original logs and index into a zip file. The program must also support a search mode that loads the pickle index and answers token queries.

CODE :

```
import os
import re
import pickle
import zipfile
TOKEN_PATTERN = re.compile(r"[A-Za-z0-9_]+")


def tokenize(line):


    return [
        token.lower()
        for token in TOKEN_PATTERN.findall(line)
    ]
def build_index(folder_path, zip_name):
    if not os.path.isdir(folder_path):
        print("Invalid folder path.")
        return
    index = {}

    total_files = 0
    total_lines = 0
    for filename in os.listdir(folder_path):
        full_path = os.path.join(
            folder_path,
            filename
        )

        if not os.path.isfile(full_path):
            continue

        total_files += 1

        try:
```

```
        with open(
            full_path,
            "r",
            encoding="utf-8",
            errors="ignore"
        ) as file:

            for line_number, line in enumerate(
                file,
                start=1
            ):

                total_lines += 1

                tokens = tokenize(line)
                for token in set(tokens):

                    if token not in index:
                        index[token] = []

                    index[token].append(
                        (filename, line_number)
                    )

    except
        OSError:
        continue

pickle_path = os.path.join(
    folder_path,
    "index.pkl"
)

with open(pickle_path, "wb") as file:
    pickle.dump(index, file)

with zipfile.ZipFile(
    zip_name,
    "w",
    zipfile.ZIP_DEFLATED
) as archive:
    for filename in os.listdir(folder_path):
        full_path = os.path.join(
```

```
            folder_path,
            filename
        )

        if os.path.isfile(full_path):

            if os.path.abspath(full_path) == os.path.abspath(
                zip_name
            ):
                continue

            archive.write(
                full_path,
                arcname=filename
            )
    print(f"FILES {total_files}")
    print(f"LINES {total_lines}")
    print(f"TOKENS {len(index)}")

def search_index(pickle_path, queries):
    try:
        with open(pickle_path, "rb") as file:
            index = pickle.load(file)

    except FileNotFoundError:
        print("Pickle index not found.")
        return

    for query in queries:
        token = query.lower()
        matches = index.get(token, [])
        print(f"{token}:")

        for filename, line_number in matches:
            print(f"{filename}:{line_number}")

def main():
    mode = input().strip().upper()
    if mode == "BUILD":
        parts = input().split()


        if len(parts) != 2:
```

```
        print("Invalid BUILD format.")return

    folder_path = parts[0]
    zip_name = parts[1]

    build_index(
        folder_path,
        zip_name
    )
elif mode == "SEARCH":
    parts = input().split()

    if len(parts) < 2:
        print("Invalid SEARCH format.")
        return
    pickle_path = parts[0]
    q = int(parts[1])
    queries = []

    for _ in range(q):
        queries.append(input().strip())

    search_index(
        pickle_path,
        queries
    )
else:
    print("Invalid mode. Use BUILD or
SEARCH.") if __name__ == "__main__":
main()
```

OUTPUT:

```
BUILD
logs archive.zip
FILES 3
LINES 9
TOKENS 13
```

PROBLEM STATEMENT 9: Simulate a job scheduler where each job has job_id, priority, duration and required resource count. Multiple worker threads execute jobs. Higher priority jobs must be processed first; jobs with same priority use earlier arrival order. The program must maintain thread-safe job assignment and produce a final execution report.

CODE :

```
import heapq
import threading
from dataclasses import dataclass
@dataclass
class Job:
    arrival: int
    job_id: str
    priority: int
    duration: int
    resources: int
    order: int
def simulate(jobs, worker_count):

    jobs.sort(
        key=lambda job: (
            job.arrival,
            job.order
        )
    )
     waiting = []
     workers = [
         (0, i + 1)
        for i in range(worker_count)
    ]
    heapq.heapify(workers)
    results = []
    index = 0
    current_time = 0
    while index < len(jobs) or waiting:

        if not waiting and index < len(jobs):
            current_time = max(
                current_time,
                jobs[index].arrival
            )

        while (
```

```
    index < len(jobs)and jobs[index].arrival <= current_time
):
    job = jobs[index]
    heapq.heappush(
        waiting,
        (
            -job.priority,
            job.arrival,
            job.order,
            job
        )
    )

    index += 1
finish_time, worker_id = heapq.heappop(
    workers
)
current_time = max(
    current_time,
    finish_time
)
while (
    index < len(jobs)
    and jobs[index].arrival <= current_time
):
    job = jobs[index]
    heapq.heappush(
        waiting,
        (
            -job.priority,
            job.arrival,
            job.order,
            job
        )
    )

    index += 1
if not waiting:
    heapq.heappush(
        workers,
        (current_time, worker_id)
    )
```

```
            continue
        _, _, _, job = heapq.heappop(waiting)

        start_time = max(
            current_time,
            job.arrival,
            finish_time
        )

        finish_time  =
            (  start_time
            +
            job.duration
        )

        waiting_time =
            ( start_time -
            job.arrival
        )
        results.append(
            (
                job.job_id,
                worker_id,
                start_time,
                finish_time,
                waiting_time
            )
        )

        heapq.heappush(
            workers,
            (
                finish_time,
                worker_id
            )
        )
        current_time = start_time
    return results
def main():
    w, n = map(int, input().split())
    if not (1 <= w <= 64):
        raise ValueError("Workers must be between 1 and 64.")
```

```
        jobs = []
    for order in range(n):
        arrival, job_id, priority, duration, resources = (input().split()
        )
        job = Job(
            int(arrival),
            job_id,
            int(priority),
            int(duration),
            int(resources),
            order
        )
        jobs.append(job)
    results = simulate(
        jobs,
        w
    )
    total_wait = sum(
        result[4]
        for result in results
    )
    for job_id, worker_id, start, finish, wait in results:
        print(
            f"{job_id} W{worker_id} "
            f"{start} {finish}"
     )
    average_wait = (
        total_wait / len(results)
        if results
        else 0
    )
    print(
        f"AVG_WAIT {average_wait:.2f}"
    )
if __name__ == "__main__":
    main()
```

OUTPUT:

```
2 3
0 J1 2 5 1
1 J2 5 3 1
2 J3 2 2 1
J1 W1 0 5
J2 W2 1 4
J3 W2 4 6
AVG_WAIT 0.67
```

PROBLEM STATEMENT 10: Create a Tkinter GUI application to manage student assignment submissions. The application must support adding a student, adding a submission, updating marks, filtering by pending/completed status, and exporting a CSV report. Data must persist locally using JSON or CSV files. Input fields must be validated before saving.

CODE :

```
import tkinter as tk
from tkinter import ttk, messagebox, filedialog
import json
import csv
import os

DATA_FILE = "assignments.json"

class AssignmentTracker:

    def _init_(self, root):

        self.root = root
        self.root.title("Assignment Tracker")
        self.root.geometry("900x600")

        self.records = self.load_data()
        self.create_widgets()

        self.refresh_table()

    def load_data(self):

        if not os.path.exists(DATA_FILE):
            return []

        try:

            with open(
                DATA_FILE,
                "r",
                encoding="utf-8"
            ) as file:

                return json.load(file)

        except (json.JSONDecodeError, OSError):
```

```
            return []
    def save_data(self):
        with open(
            DATA_FILE,
            "w",
            encoding="utf-8"
        ) as file:

            json.dump(
                self.records,
                file,
                indent=4
            )
    def create_widgets(self):
        title = tk.Label(
            self.root,
            text="Student Assignment Tracker",
            font=("Arial", 18, "bold")
        )
        title.pack(pady=10)

        input_frame = tk.Frame(self.root)
        input_frame.pack(pady=5)

        tk.Label(
            input_frame,
            text="Enrollment:"
        ).grid(row=0, column=0, padx=5)

        self.enrollment_entry = tk.Entry(
            input_frame
        )

        self.enrollment_entry.grid(
            row=0,
            column=1,
            padx=5
        )

        tk.Label(
```

```
        input_frame,
        text="Name:"
    ).grid(row=0, column=2, padx=5)

    self.name_entry = tk.Entry(
        input_frame
    )

    self.name_entry.grid(
        row=0,
        column=3,
        padx=5
    )

    tk.Label(
        input_frame,
        text="Assignment:
        "
    ).grid(row=1, column=0, padx=5)

    self.assignment_entry = tk.Entry(
        input_frame
    )

    self.assignment_entry.grid(
        row=1,
        column=1,
        padx=5
    )

    tk.Label(
        input_frame,
        text="Status:"
    ).grid(row=1, column=2, padx=5)

    self.status_combo = ttk.Combobox(
        input_frame,
        values=["Pending", "Completed"],
        state="readonly"
    )
    self.status_combo.set("Pending")


    self.status_combo.grid(
```

```
        row=1,column=3, padx=5
    )

    tk.Label(
        input_frame,
        text="Marks:"
    ).grid(row=2, column=0, padx=5)

    self.marks_entry = tk.Entry(
        input_frame
    )

    self.marks_entry.grid(
        row=2,
        column=1,
        padx=5
    )

    tk.Label(
        input_frame,
        text="Remarks:"
    ).grid(row=2, column=2, padx=5)

    self.remarks_entry = tk.Entry(
        input_frame
    )

    self.remarks_entry.grid(
        row=2,
        column=3,
        padx=5
    )

    button_frame = tk.Frame(self.root)
    button_frame.pack(pady=10)

    tk.Button(
        button_frame,
        text="Add Submission",
        command=self.add_submission
    ).grid(row=0, column=0, padx=5)

    tk.Button(
```

```
    button_frame,
    text="Update Marks",
    command=self.update_marks
).grid(row=0, column=1, padx=5)

tk.Button(
    button_frame,
    text="Export CSV",
    command=self.export_csv
).grid(row=0, column=2, padx=5)

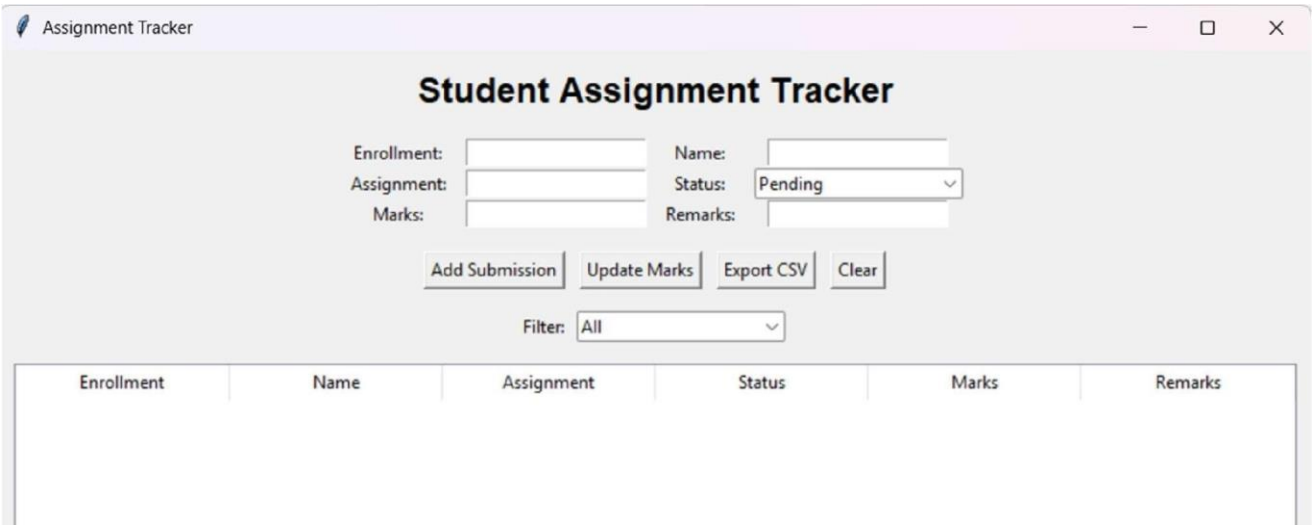
tk.Button(
    button_frame,
    text="Clear",
    command=self.clear_fields
).grid(row=0, column=3, padx=5)

filter_frame = tk.Frame(self.root)
filter_frame.pack(pady=5)

tk.Label(
    filter_frame,
    text="Filter:"
).pack(side=tk.LEFT)

self.filter_combo = ttk.Combobox(
    filter_frame,
    values=[
        "All",
        "Pending",
        "Completed"
    ],
    state="readonly"
)
self.filter_combo.set("All")
self.filter_combo.pack(
    side=tk.LEFT,
    padx=5
)
```

OUTPUT:


Assignment Tracker
Student Assignment Tracker
Enrollment:
Name:
Assignment:
Status:
Pending
Marks:
Remarks:
Add Submission
Update Marks
Export CSV
Clear
Filter:
All
Enrollment
Name
Assignment
Status
Marks
Remarks